- 三角函数有理式积分.

$$\int R(\sin x, \cos x)\,dx$$

$$R(\sin x, \cos x) = \frac{P(\sin x, \cos x)}{Q(\sin x, \cos x)}$$

令 $\tan\frac{x}{2} = t$，则

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$x = 2\arctan t \ (+2k\pi), \quad dx = \frac{2\,dt}{1+t^2}$$

于是

$\int R(\sin x, \cos x)\,dx$ 化为 t 的有理函数积分.

例：$\int \frac{dx}{2\sin x-\cos x+5}$

解：令 $\tan\frac{x}{2}=t$，则

$$\text{原式}=\int \frac{1}{2\cdot\frac{2t}{1+t^2}-\frac{1-t^2}{1+t^2}+5}\cdot\frac{2dt}{1+t^2}$$

$$=\int \frac{dt}{3t^2+2t+2}=\frac{1}{3}\int \frac{d(t+\frac{1}{3})}{(t+\frac{1}{3})^2+(\frac{\sqrt{5}}{3})^2}$$

$$=\frac{1}{3}\cdot\frac{3}{\sqrt{5}}\cdot\arctan\frac{3}{\sqrt{5}}(t+\frac{1}{3})+C$$

$$=\frac{1}{\sqrt{5}}\arctan\frac{3t+1}{\sqrt{5}}+C$$

$$=\frac{1}{\sqrt{5}}\arctan\frac{3\tan\frac{x}{2}+1}{\sqrt{5}}+C$$

- 不是所有的有理三角函数都要那样代换.

例: $\int \frac{\sin^3 x}{1+\cos x}\,dx$

$= \int \frac{\sin^2 x}{1+\cos x}\cdot \sin x\,dx$

$= -\int \frac{1-\cos^2 x}{1+\cos x}\cdot d(\cos x)$

$= -\int (1-\cos x)\,d(\cos x)$

$= -\cos x + \frac{1}{2}\cos^2 x + C$

☆ 如果用万能变换

令 $\tan\frac{x}{2}=t$, 则

原式 $= \int \frac{\left(\frac{2t}{1+t^2}\right)^3}{1+\frac{1-t^2}{1+t^2}}\cdot\frac{2\,dt}{1+t^2} = \int \frac{8t^3}{(1+t^2)^3}\,dt$

(比上面做法复杂)

例: $\int \sec^3 x\,dx$ (见例 2.14)

解法1: 令 $\tan\frac{x}{2}=t$, 则

$$\sec x=\frac{1}{\cos x}=\frac{1+t^2}{1-t^2}$$

$$\int \sec^3 x\,dx=\int\left(\frac{1+t^2}{1-t^2}\right)^3\cdot\frac{2\,dt}{1+t^2}$$

$$=2\int\frac{(1+t^2)^2}{(1-t^2)^3}\,dt$$

利用 $(1+t^2)^2=(1-t^2)^2+4(t^2-1)+4$

可算出结果.

解法2: $\int \sec^3 x\,dx=\int\frac{\cos x\,dx}{\cos^4 x}$

$$=\int\frac{d(\sin x)}{(1-\sin^2 x)^2}\xrightarrow{u=\sin x}\int\frac{du}{(1-u^2)^2}$$

设 $\frac{1}{(1-u^2)^2}=\frac{A}{1-u}+\frac{B}{(1-u)^2}+\frac{C}{1+u}+\frac{D}{(1+u)^2}$

对于A.B.C.D进行求有理积分.

- 例: $\int \frac{\sin^2 x}{1+\cos^2 x}\,dx$

解: 令 $\tan x = t$, 则

$$\cos^2 x = \frac{1}{\sec^2 x} = \frac{1}{1+\tan^2 x} = \frac{1}{1+t^2}$$

$$\sin^2 x = 1-\cos^2 x = \frac{t^2}{1+t^2}$$

$$x = \arctan t, \quad dx = \frac{dt}{1+t^2}$$

$$\text{原式} = \int \frac{\frac{t^2}{1+t^2}}{1+\frac{1}{1+t^2}} \cdot \frac{dt}{1+t^2}$$

$$= \int \frac{t^2}{(2+t^2)(1+t^2)}\,dt$$

$$= \int \frac{dt}{2+t^2} - \int \frac{1}{(2+t^2)(1+t^2)}\,dt$$

$$= \int \frac{dt}{2+t^2} - \int \left(\frac{1}{1+t^2} - \frac{1}{2+t^2}\right) dt$$

$$= 2\int \frac{dt}{2+t^2} - \int \frac{dt}{1+t^2}$$

$$= \frac{2}{\sqrt{2}} \cdot \arctan \frac{t}{\sqrt{2}} - \arctan t + C$$

$$= \sqrt{2} \arctan\left(\frac{\tan x}{\sqrt{2}}\right) - x + C$$

☆ 若令 $\tan \frac{x}{2} = t$, 则

$$\text{原式} = \int \frac{\left(\frac{2t}{1+t^2}\right)^2}{1+\left(\frac{1-t^2}{1+t^2}\right)^2} \cdot \frac{2dt}{1+t^2}$$

比较难算.

☆ 其它解法见书.

- 一类无理函数的积分：

$$\int R\left(x, \sqrt[n]{\frac{Ax+B}{Cx+D}}\right) dx .$$

令 $\sqrt[n]{\frac{Ax+B}{Cx+D}} = t$，则 $\frac{Ax+B}{Cx+D} = t^n$.

反解出来 $x = \frac{Et^n+F}{Gt^n+H}$.

$dx = \frac{\cdots}{\cdots} dt$,

从而原积分可化为有理函数积分.

- 例：$\int \frac{\sqrt{x+2}}{x+3}\,dx$

解：设 $\sqrt{x+2}=t$，则 $x+2=t^2$

$x=t^2-2$，$dx=2t\,dt$

$$\text{原式}=\int \frac{t}{t^2+1}\cdot 2t\,dt$$

$$=2\int\left(1-\frac{1}{t^2+1}\right)dt$$

$$=2t-2\arctan t$$

$$=2\sqrt{x+2}-2\arctan\sqrt{x+2}+C$$

• 例：$\int \frac{1}{x}\cdot\sqrt{\frac{x+2}{x-2}}\,dx$

解：设 $\sqrt{\frac{x+2}{x-2}}=t$，则 $\frac{x+2}{x-2}=t^2$

$$x=\frac{2(t^2+1)}{t^2-1},\quad dx=\frac{-8t}{(t^2-1)^2}dt$$

$$\text{原式}=\int\frac{t^2-1}{2(t^2+1)}\cdot t\cdot\frac{-8t}{(t^2-1)^2}dt$$

$$=\int\frac{-4t^2}{(t^2+1)(t^2-1)}dt$$

$$=-2\int\left(\frac{1}{t^2+1}+\frac{1}{t^2-1}\right)dt$$

$$=-2\arctan t+\ln\left|\frac{t+1}{t-1}\right|+C$$

$$= -2\arctan\sqrt{\frac{x+2}{x-2}} + \ln\left|\frac{1+\sqrt{\frac{x+2}{x-2}}}{1-\sqrt{\frac{x+2}{x-2}}}\right| + C$$

☆这里 $x>2$ 或者 $x\leq -2$. 如果化简需要注意. $(x\neq 0)$

- 例: $\int \frac{dx}{\sqrt[4]{(x-2)^3(x+1)^5}}$

解: 这里 $x<-1$ 或 $x>2$.

$$\frac{1}{\sqrt[4]{(x-2)^3(x+1)^5}} = \frac{1}{(x+1)^2}\cdot\sqrt[4]{\left(\frac{x+1}{x-2}\right)^3}$$

设 $\sqrt[4]{\frac{x+1}{x-2}} = t$. 则

$$\frac{x+1}{x-2} = t^4, \quad x = \frac{2t^4+1}{t^4-1}$$

$$dx = \frac{-12t^3}{(t^4-1)^2}dt$$

$$\text{原式} = \int \frac{1}{\left(\frac{2t^4+1}{t^4-1}+1\right)^2} \cdot t^3 \cdot \frac{-12t^3}{(t^4-1)^2}dt$$

$$= \int \frac{-12t^6}{(3t^4)^2}dt = -\frac{4}{3}\int \frac{dt}{t^2}$$

$$= \frac{4}{3t} + C = \frac{4}{3} \cdot \sqrt[4]{\frac{x-2}{x+1}} + C$$

- 没有绝对的万能积分法
- 选择简单的适合自己的方法

作业：

习题4-4A类

1. (3). (5).

2. (1)

习题4-4B类

2. (1). (2)